

CHIP-2070: Prototyping Edition

3D Micro-Fabrication Blueprint in Fused Silica via Femtosecond Laser Writing

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2026

Abstract

This technical specification document establishes the formal, geometrically validated 3D architecture for the CHIP-2070 micro-prototype. Fabricated within a fused silica (SiO_2) substrate utilizing Femtosecond Laser Writing (FLW), the architecture implements an intensity-dependent nonlinear optical switching mechanism (“Boghian-Shift”). This is achieved via asymmetric evanescent coupling across a multi-layered waveguide grid. This document provides precise nanometer-scale XYZ coordinate maps, a mathematically verified 42.00° nonlinear switching angle, localized Kerr gate topologies, automated geometric validation protocols, and machine-ready pseudo G-code for direct laboratory fabrication.

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1 Technical Specification Parameters

The physical, structural, and electro-optical operational parameters required for the baseline calibration of the FLW system and detection hardware are specified in Table 1.

Table 1: CHIP-2070 System and Fabrication Parameters

Parameter Group	Value	Technical Notes / Calibration
Substrate Properties		
Substrate Material	Fused Silica (SiO_2)	High-purity, optical insertion loss < 0.1 dB/cm
Chip Dimensions	$10 \times 10 \times 1$ mm	Standard polished die for FLW mounting
FLW Laser Engine		
Laser Wavelength	1030 nm	Ytterbium-doped fiber laser source
Pulse Duration	100 fs	Compressed pulse width at target
Pulse Energy Range	$5 - 20 \mu\text{J}$	Tunable threshold for local modification
Objective Numerical Aperture	$0.4 - 0.65$	Optimized for localized 3D focal spots
Waveguide Geometry		
Waveguide Core Diameter	$6 - 8 \mu\text{m}$	Single-mode operational profile at 1030 nm
Structural Refractive Index (Δn)	$+1.5 \times 10^{-3}$	Parabolic gradient profile in center of core
Target Coupling Gap	$2.0 - 3.0 \mu\text{m}$	Evanescent tunneling zone proximity
Kerr Interaction Length	$80 \mu\text{m}$	Effective length of the nonlinear gate
Functional 3D Angle (θ)	42.00°	Strictly validated Total Internal Reflection (TIR)
Layer Distribution	2 Layers	Stratified at $Z = -100 \mu\text{m}$ and $Z = -150 \mu\text{m}$
Inter-channel Cross-talk	< -30 dB	Equivalent to $\leq 0.1\%$ parasitic leakage
Detection Hardware		
Photodetector Array	InGaAs Pin	High-speed transient response < 50 ps rise time
Sampling Oscilloscope	≥ 33 GHz	Real-time single-shot pulse tracking
Signal Processing	Lock-In (SR830)	Phase-locked to laser repetition frequency

2 3D Architecture and Stratification Maps

The coordinate system defines the origin (0,0,0) at the front-left-top corner of the chip. The Z-axis extends vertically downward into the bulk substrate (positive depth values indicated as negative coordinates relative to the surface plane).

2.1 Layer 1: Primary Core Network ($Z = -100 \mu\text{m}$)

Layer 1 hosts the main operational matrix. Channels are laid out with a constant lateral pitch of $30 \mu\text{m}$ on the Y-axis to prevent passive cross-talk.

Table 2: Layer 1 Spatial Coordinate Assignments

Waveguide Identifier	X Coordinate (μm)	Y Coordinate (μm)	Z Coordinate (μm)
Core 1	100.00	200.00	-100.00
Core 2	100.00	230.00	-100.00
Core 3 (Active Injection)	100.00	260.00	-100.00
Core 4	100.00	290.00	-100.00
Core 5	100.00	320.00	-100.00

2.2 Layer 2: Energy Dissipation Network ($Z = -150 \mu\text{m}$)

Layer 2 functions as the structural sink (Dump) layer, situated exactly $50 \mu\text{m}$ below Layer 1 to isolate high-intensity transients.

Table 3: Layer 2 Spatial Coordinate Assignments

Waveguide Identifier	X Coordinate (μm)	Y Coordinate (μm)	Z Coordinate (μm)
Dump 1	300.00	200.00	-150.00
Dump 2	300.00	230.00	-150.00
Dump 3 (Active Sink)	300.00	260.00	-150.00
Dump 4	300.00	290.00	-150.00
Dump 5	300.00	320.00	-150.00

—

3 Kerr Gate Geometry and Trigonometric Correction

To achieve the precise intensity-dependent nonlinear switching threshold (the Boghian-Shift effect) at an exact physical angle of 42.00° inside the XZ plane, the path coordinates must account for the refractive boundaries. A raw linear interpolation would deform the wavefront.

3.1 Segment 1: Core 3 Extraction Tail

Linear extension from the primary routing layer to stabilize the single-mode profile before entering the switching junction.

- **Junction Launch Point** (P_{S1_Start}): (100.00, 260.00, -100.00)
- **Junction Transition Point** (P_{S1_End}): (130.00, 260.00, -110.00)

3.2 Segment 2: Corrected 42.00° Nonlinear Diagonal

The geometric offset requires exactly $35 \mu\text{m}$ of vertical translation along the Z-axis (from $Z = -110 \mu\text{m}$ to $Z = -145 \mu\text{m}$). To enforce a strict structural slope ($\theta = 42.00^\circ$), the horizontal displacement (ΔX) is explicitly constrained via:

$$\Delta X = \frac{|\Delta Z|}{\tan(42.00^\circ)} = \frac{35.00 \mu\text{m}}{0.900404} = 38.8718 \mu\text{m} \approx 38.87 \mu\text{m} \quad (1)$$

This mathematical derivation yields the absolute nanometer-calibrated boundary points:

- **Diagonal Entry Point** (P_{S2_Start}): (130.00, 260.00, -110.00)
- **Diagonal Exit Point** (P_{S2_End}): (168.87, 260.00, -145.00)

3.3 Segment 3: Dump 3 Coupling Entry

Final horizontal convergence interface designed to guide the switched mode cleanly into the dissipation layer.

- **Sink Interface Start** (P_{S3_Start}): (168.87, 260.00, -145.00)
- **Sink Capture Target** (P_{S3_End}): (300.00, 260.00, -150.00)

—

4 Sample Preparation and Experimental Tolerance Requirements

To ensure the successful physical execution of this 3D blueprint across different laboratory FLW setups, the fabrication facility should note the following operational flexibilities and technical requirements:

4.1 Refractive Index Refraction Correction (Z -Scan)

The nominal depths specified in this document ($Z = -100 \mu\text{m}$ and $Z = -150 \mu\text{m}$) represent targeted focal positions inside the bulk glass. The fabrication engineer must apply the appropriate geometric correction factor to compensate for the refractive index mismatch between air ($n \approx 1.0$) and Fused Silica ($n \approx 1.45$ at 1030 nm). The stage translation software must adjust the physical Z -axis depth ($Z_{\text{actual}} = Z_{\text{nominal}} \cdot \frac{n_{\text{glass}}}{n_{\text{air}}}$) to prevent spherical aberration and guarantee that the actual written layers are perfectly flat and separated by exactly $50.00 \mu\text{m}$.

4.2 Dynamic Coupling Gap Optimization

While the target evanescent coupling gap is mathematically locked at $2.5 \mu\text{m}$, the final cross-talk efficiency depends heavily on the laser's voxel shape (determined by the objective's NA and plasma formation threshold). The fabrication facility is encouraged to generate a sample matrix using a parameter sweep:

- **Allowable Gap Variance:** $2.0 \mu\text{m}$ to $3.0 \mu\text{m}$, mapped in increments of $0.2 \mu\text{m}$ across separate test dies.
- **Threshold Calibration:** If the written waveguide exhibits higher intrinsic Δn than $+1.5 \times 10^{-3}$, the gap should be widened toward $3.0 \mu\text{m}$ to avoid passive directional coupling in the low-intensity regime.

4.3 Anisotropic Thermal Stress Dissipation

To prevent micro-fractures in the glass matrix during high-speed, continuous-wave voxel translation, all 3D curved paths (Segment 1, 2, and 3) must be written using a multi-pass technique or a reduced feed rate ($\leq 20 \mu\text{m/s}$). A post-fabrication thermal annealing process (650°C for 2 hours followed by a slow ramp-down to room temperature) is highly recommended to relieve localized material stresses around the 42.00° Kerr gating junction.

5 Geometric Angle Verification Protocol

To automate pre-fabrication checks in the CNC software suite, the system evaluates the real physical slope angle θ using the native coordinates of Segment 2:

$$\theta = \arctan\left(\frac{|Z_2 - Z_1|}{|X_2 - X_1|}\right) = \arctan\left(\frac{|-145.00 - (-110.00)|}{|168.87 - 130.00|}\right) = \arctan\left(\frac{35.00}{38.87}\right) \equiv 42.0016^\circ \quad (2)$$

The structural acceptance tolerance for the laser system software is defined as:

$$|\theta_{\text{measured}} - 42.00^\circ| \leq 0.05^\circ \quad (3)$$

5.1 Automated Validation Routine (Pseudocode)

The following algorithmic function is executed by the compiler prior to engaging the laser gating systems:

```

function validate_boghian_angle(P1, P2):
    dx = abs(P2.x - P1.x)
    dz = abs(P2.z - P1.z)
    if dx == 0:
        return "INVALID_ZERO_DIVISION"

    theta = atan(dz / dx) * 180.0 / pi
    error_margin = abs(theta - 42.00)

    if error_margin <= 0.05:
        return "VALIDATION_PASSED [ANGLE = " + toString(theta) + " deg]"
    else:
        return "VALIDATION_FAILED [ANGLE = " + toString(theta) + " deg]"

```

6 Structural Topology Visualization

The structural configuration map below shows the relationship between the single-mode tracking channels and the asynchronous 3D Kerr switching diagonal.

0.2cm

```

[LATERAL PROJECTION VIEW - XZ PLANE]

Surface (Z = 0) -----

Layer 1 (Z=-100 um) -> [Core 3]--* (130, 260, -110)
                        \
                        \ Asymmetric Evanescent Coupling Zone
                        \ [True 42.00 deg Gradient Slope]
                        \
Layer 2 (Z=-150 um) -> *---(168.87, 260, -145)-----[Dump 3]
-----

```

7 Machine-Ready Pseudo G-Code for FLW Systems

The instructional scripting logic blocks in this section provide the raw operational paths for multi-axis Aerotech or Newport translation stages equipped with fast-switching AOM (Acousto-Optic Modulator) laser shutters.

```

; =====
; SYSTEM INITIALIZATION & GLOBAL PARAMETER LOAD
; =====
SET_LASER_WAVELENGTH 1030nm
SET_PULSE_DURATION 100fs
SET_PULSE_ENERGY 10uJ
SET_REPETITION_RATE 5kHz
SET_OBJECTIVE_NA 0.50
SET_FEEDRATE_UNITS um_per_sec

; =====
; SUB-ROUTINE DEFINITION FOR PARAMETRIC WAVEGUIDE ETCHING
; =====
DEFINE_WAVEGUIDE_PATH(name, x1, y1, z1, x2, y2, z2, linear_speed):
    COMMAND_MOVE_XYZ(x1, y1, z1)
    SHUTTER_AOM_ENGAGE(STATE=ON)

```

```

    COMMAND_FEED_XYZ(x2, y2, z2, SPEED=linear_speed)
    SHUTTER_AOM_ENGAGE (STATE=OFF)
END_DEFINE

; =====
; FABRICATION SEQUENCE EXECUTION
; =====
; 1. Inscribing Layer 1 Reference Waveguide (Core 3 Active)
EXECUTE DEFINE_WAVEGUIDE_PATH(CORE3, 100.00, 260.00, -100.00, 1000.00, 260.00,
    -100.00, 50)

; 2. Inscribing Layer 2 Reference Waveguide (Dump 3 Sink)
EXECUTE DEFINE_WAVEGUIDE_PATH(DUMP3, 300.00, 260.00, -150.00, 1200.00, 260.00,
    -150.00, 50)

; 3. Inscribing High-Precision Kerr Gate Junction Segments
EXECUTE DEFINE_WAVEGUIDE_PATH(SEG1_LAUNCH, 100.00, 260.00, -100.00, 130.00,
    260.00, -110.00, 20)
EXECUTE DEFINE_WAVEGUIDE_PATH(SEG2_DIAGONAL, 130.00, 260.00, -110.00, 168.87,
    260.00, -145.00, 20)
EXECUTE DEFINE_WAVEGUIDE_PATH(SEG3_CAPTURE, 168.87, 260.00, -145.00, 300.00,
    260.00, -150.00, 20)

; =====
; SYSTEM SHUTDOWN PROCEDURES
; =====
SHUTTER_AOM_ENGAGE (STATE=OFF)
RETURN_HOME_XYZ(0, 0, 0)
SYSTEM_E_STOP_READY

```

8 Conclusion and Verification Sign-Off

This technical manual outlines the complete spatial configuration, mathematical geometry, evaluation criteria, and execution code needed to manufacture a functional CHIP-2070 micro-prototype. By updating the horizontal steps to match the target refractive benchmarks, the design guarantees a uniform 42.00° nonlinear path. This prevents leakage across channels and provides a reliable setup for multi-layer optical validation tests.

CHIP-2070 Design Authority
Verified for Physical Laser Prototyping Infrastructure (Cycles 2026–2027)